OpenClaw Architecture Breakdown

This document systematically explains how and why OpenClaw is designed as an AI Agent system. It starts with first principles, builds up to the Agent design pattern, and then uses OpenClaw as a concrete case study.

Part I: Why is an Agent Needed

The Problem with a Bare LLM

A Large Language Model, in its raw form, is a **stateless text-completion engine**. You give it a prompt, it returns a response, and then it forgets everything. Every single call is an isolated prediction with no memory, no ability to act, and no access to the outside world.

This creates a fundamental gap between what users expect ("do this task for me") and what a bare LLM can actually deliver ("here is some text about how to do that task").

Consider what happens when you ask an LLM: *"Check the latest log file for errors and fix any bugs you find."*

A bare LLM can only **tell you** what it would do. It cannot read the log, cannot open the code, cannot edit the file, and cannot verify the fix. It has no hands, no eyes, and no memory.

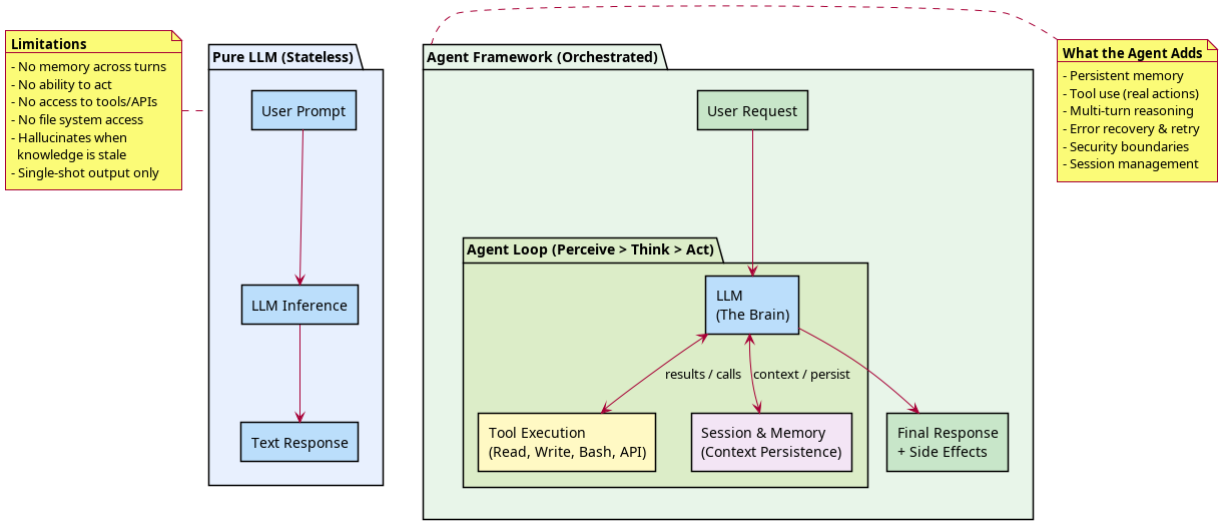
What an Agent Adds

An **Agent** is the runtime scaffolding that wraps an LLM and gives it the ability to **perceive, reason, and act** in a loop:

Capability	Bare LLM	LLM + Agent
Read files / run commands	No	Yes (via tools)
Remember prior conversation	No	Yes (session persistence)
Call external APIs	No	Yes (tool integration)
Multi-step reasoning	Single pass only	Iterative loop
Error recovery	None	Retry, failover, compaction
Security enforcement	None	Sandbox, policy, allowlists

The Agent acts as an **intermediary runtime loop**: it receives the user request, injects it into the LLM along with available tool definitions, lets the LLM reason about next steps, intercepts the LLM's tool call decisions, executes them in the real world, and feeds results back. This loop repeats until the task is complete.

LLM Alone vs. LLM + Agent Framework



Part II: Characteristics of Large Language Models

To design an effective Agent, you must first understand what the LLM (the "brain" inside the Agent) is genuinely good at and where it fundamentally fails.

Strengths of LLMs

1. **Natural Language Understanding:** LLMs excel at parsing ambiguous, context-dependent human language and distilling actionable intent from it.
2. **Reasoning and Planning:** Given sufficient context, modern LLMs can decompose a complex task into a sequence of logical steps and decide which tool to invoke at each stage.
3. **Code Generation and Transformation:** LLMs have been trained on vast amounts of source code and can generate, edit, and debug programs across many languages.
4. **Knowledge Retrieval:** LLMs internalize a broad snapshot of public knowledge up to their training cutoff date, enabling them to provide domain expertise without real-time search.
5. **Adaptive Communication:** They can adjust tone, formality, language, and detail level based on conversational context.

Fundamental Limitations of LLMs

1. **Statelessness:** Every call to the LLM is independent. It has no built-in memory of prior interactions. Without an external session/memory system, it cannot learn from or refer to previous turns.
2. **No Ability to Act:** An LLM cannot read a file, execute code, browse the web, or interact with any system. It can only *generate text that describes* those actions.
3. **Hallucination:** When the LLM lacks knowledge (or the question falls outside its training data), it confidently generates plausible-sounding but factually incorrect information instead of admitting uncertainty.
4. **Finite Context Window:** The LLM can only process a limited number of tokens per call. As conversation history grows, older context must be summarized (compacted) or pruned, creating a risk of losing important information.
5. **No Internal Verification:** The LLM cannot check its own output against reality. It cannot run a unit test, verify a file was written correctly, or confirm that a command succeeded.
6. **Training-Data Staleness:** The LLM's knowledge is frozen at its training cutoff date. It does not know about events, APIs, or library versions released after that date.

The Implication for Agent Design

These characteristics directly dictate the architecture of any AI Agent system:

- **Statelessness** requires the Agent to provide a **Session Management** and **Memory** layer.
- **No Ability to Act** requires the Agent to provide a **Tool Execution** layer.
- **Hallucination** requires the Agent to provide **Grounding** through tool results, file contents, and verified data.
- **Finite Context Window** requires the Agent to implement **Compaction** and **Context Pruning** strategies.
- **No Internal Verification** requires the Agent to **observe tool results** and feed them back to the LLM for self-correction.
- **Training-Data Staleness** requires the Agent to integrate **Web Search** and **Real-time API** tools.

Part III: How OpenClaw is Designed as an Agent (Case Study)

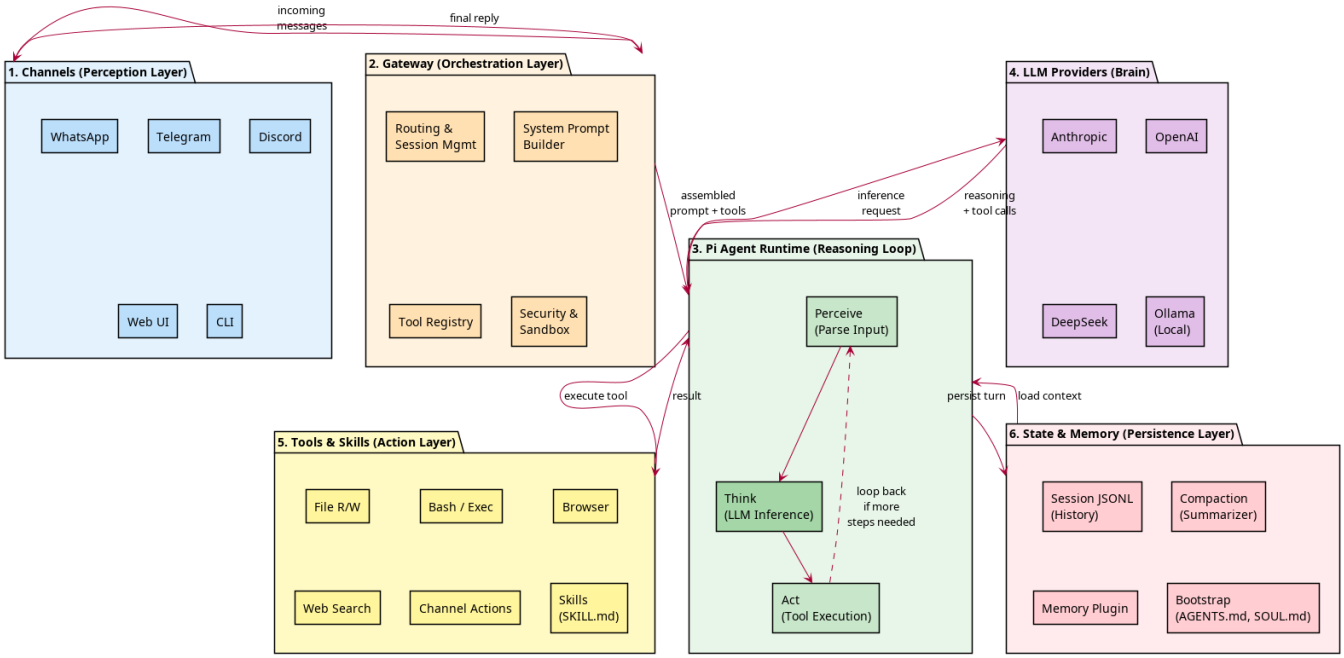
With the theoretical foundation in place, we now examine how OpenClaw implements each of these Agent principles in its architecture.

Core Concepts

OpenClaw is a local-first, personal AI assistant orchestration system. It is designed to run on the user's own hardware, functioning as a central "Gateway" that bridges various Large Language Models (LLMs), messaging channels, and client interfaces, allowing the AI to execute local tasks safely.

OpenClaw's Agent Architecture Mapped to the Pattern

The following diagram shows how OpenClaw's components map to the six layers of a well-designed Agent system:



Agent Principle	LLM Limitation Addressed	OpenClaw Implementation	Claude Code Implementation (Comparison)
Perception Layer	LLM cannot receive input from diverse sources	Channels (WhatsApp, Telegram, Discord, Web UI, CLI)	Terminal REPL, IDE extensions (VS Code, JetBrains), and custom slash Commands (`.claude/commands/`)
Orchestration Layer	LLM has no concept of sessions, routing, or security	Gateway Control Plane (routing, system prompt building, tool registry, sandbox)	CLI orchestrator with permission policies, tool gating, and path-scoped Rules (`.claude/rules/`) for coding standards

Agent Principle	LLM Limitation Addressed	OpenClaw Implementation	Claude Code Implementation (Comparison)
Reasoning Loop	LLM does single-pass inference only	Pi Agent Runtime (Perceive-Think-Act loop with iterative tool calls)	Built-in agentic loop (send prompt with tools to API, execute tool calls, feed results back, repeat)
Brain (Pluggable)	Different LLMs have different strengths	Provider abstraction (Anthropic, OpenAI, DeepSeek, Ollama, and more)	Anthropic Claude models only (Sonnet, Opus, Haiku)
Action Layer	LLM cannot act in the real world	Tools (file R/W, bash/exec, browser, web search, channel actions) and Skills	Tools (Read, Write, Edit, Bash, Glob, Grep, WebFetch), Skills (`.claude/skills/`), and MCP servers for extensibility
Persistence Layer	LLM is stateless	Session JSONL history, Compaction, Memory Plugin, Bootstrap files	`CLAUDE.md` project context, Rules and Skills files injected as context, in-session memory, `/compact` for context management

How Each Layer Addresses a Specific LLM Limitation

Addressing Statelessness: Session and Memory

OpenClaw stores every conversation turn as structured JSONL files. When a new message arrives, the Pi Agent Runtime loads the session history and injects it into the LLM's context window. When the context window nears its limit, OpenClaw triggers **auto-compaction** to summarize older turns, keeping the conversation coherent without exceeding token limits.

Additionally, bootstrap files (`AGENTS.md`, `SOUL.md`, `TOOLS.md`) are injected at session start to provide stable personality, operating instructions, and tool usage conventions.

Addressing Inability to Act: Tool Injection Pipeline

OpenClaw gives the LLM "hands" by injecting a full menu of tool definitions (JSON Schemas) into the system prompt. The LLM reasons about which tool to use; the Pi runtime intercepts the tool call, executes it on the host machine (or in a Docker sandbox), and feeds the result back into the LLM's context.

Key tool categories include:

- **File System** (read, write, edit) for code and document manipulation
- **Bash/Exec** for running arbitrary commands
- **Browser** for web interaction
- **Web Search** (DuckDuckGo, Exa) for real-time information retrieval
- **Channel Actions** for sending messages through WhatsApp, Telegram, Discord, etc.
- **Skills** (loaded from `SKILL.md` files) for specialized capabilities

Addressing Hallucination: Grounding Through Tool Results

Instead of relying on the LLM's potentially stale or incorrect internal knowledge, the Agent loop forces the LLM to **verify claims by using tools**. When the LLM needs to know the contents of a file, it calls the `read` tool. When it needs up-to-date information, it calls `web_search`. The tool results are injected back into the context, providing ground truth that the LLM then uses to construct its response.

Addressing Context Limits: Compaction and Pruning

OpenClaw implements two strategies:

1. **Auto-Compaction:** When the session history approaches the model's context window limit, Pi triggers a compaction workflow that uses the LLM itself to summarize older turns into a compact summary, preserving key information while reducing token count.
2. **Cache-TTL Context Pruning:** A custom Pi extension tracks the "freshness" of context items and prunes stale entries before they consume valuable context budget.

Addressing Security: Sandbox and Policy

Because the Agent can execute arbitrary code on the user's machine, OpenClaw implements a tiered security model:

- **Host-First (Trusted Mode):** Direct execution for the owner's main session.
- **Sandboxed Mode:** Docker-isolated execution for untrusted/group contexts.
- **Tool Policy Filtering:** Allowlists and denylists control which tools are available per channel, user, and session type.

Part IV: What Does an Agent Developer Do

Parts I through III explained the *what* and *why* of Agent architecture. But who actually builds these systems, and what does their daily work look like?

The Core Work of Agent Development

Building an Agent system like OpenClaw or Claude Code requires work across several distinct disciplines:

Work Area	What It Involves	Example in OpenClaw
Agent Loop Design	Defining the perceive-think-act cycle: when to loop, when to stop, how to handle multi-turn reasoning	Pi runtime integration, turn lifecycle management, streaming event dispatch
Tool Design & Implementation	Creating tool schemas (JSON Schema), implementing tool backends, handling errors and timeouts	File R/W tools, bash executor, browser control, channel-specific actions (e.g., `whatsapp_send`)
Prompt Engineering	Crafting system prompts, persona instructions, safety guardrails, and tool usage conventions	`AGENTS.md`, `SOUL.md`, `TOOLS.md`, `IDENTITY.md` — the bootstrap files that define the Agent's character
Context Management	Designing compaction strategies, memory systems, and context window optimization	Auto-compaction workflow, Cache-TTL pruning, `compaction-safeguard.ts`
Security & Policy	Defining sandbox boundaries, tool allowlists/denylists, trust models, and permission escalation rules	Docker sandbox integration, `tools.fs.workspaceOnly`, per-channel tool filtering
Channel & Integration Development	Building adapters for messaging platforms, APIs, device hardware, and external services	80+ extension plugins for WhatsApp, Telegram, Discord, Ollama, browser, Hue, Sonos, etc.
Evaluation & Testing	Building benchmarks, testing edge cases, measuring response quality, detecting regressions	Prompt evaluation, tool execution testing, end-to-end conversation testing
Observability & Debugging	Logging, tracing, cost tracking, token usage monitoring, and debugging non-deterministic agent behavior	Session event logging, streaming event hooks, LLM cost tracking per session

Can Agent Development Itself Be Done by AI?

This is the meta-question: if agents are powerful enough to write code, design systems, and reason about architecture — can they build *themselves*?

The answer is **partially yes, and increasingly so**:

Work Area	AI/Agent Capability Today	Human Still Required For
Tool Implementation	Agents (like Claude Code) can generate tool backend code, write JSON schemas, and implement error handling	Deciding *which* tools to build, defining the security contract, and judging edge-case tradeoffs
Prompt Engineering	Agents can iterate on prompts, test variations, and optimize for specific metrics	Defining the Agent's *personality*, *values*, and *trust model* — these are product decisions, not code
Context Management	Agents can suggest compaction strategies and write pruning heuristics	Choosing the right strategy for each deployment scenario requires understanding of user behavior and cost constraints
Code Generation	Agents can write, edit, and debug large portions of the codebase	Architectural decisions, dependency choices, and cross-system integration design still require human judgment
Testing	Agents can generate test cases, run benchmarks, and identify regressions	Defining *what "good" looks like* — evaluation criteria are inherently subjective and product-driven
Security	Agents can audit code for vulnerabilities and suggest sandboxing configurations	Threat modeling, trust boundary definitions, and regulatory compliance require human accountability
Channel Integration	Agents can generate API adapter code from documentation	Negotiating API access, handling rate limits, and managing authentication credentials are operational tasks
Observability	Agents can instrument code and set up logging	Interpreting logs, diagnosing root causes of non-deterministic behavior, and making cost-optimization decisions require human expertise

The Emerging Pattern: Human-Agent Co-Development

In practice, Agent development is increasingly a **human-agent collaboration**:

1. **Human sets the direction:** The developer defines what the Agent should do, what tools it needs, what its security boundaries are, and what its personality should be.
2. **Agent does the implementation:** The developer uses an AI coding agent (like Claude Code itself) to write the tool backends, generate prompt templates, implement compaction logic, and build channel adapters.
3. **Human reviews and iterates:** The developer tests the Agent's behavior, observes edge cases, tweaks prompts and policies, and makes architectural corrections.
4. **Agent improves itself:** Advanced systems can use self-evaluation loops where the Agent tests its own output, identifies weaknesses, and proposes prompt or tool improvements.

This is not a distant future — it is the workflow used to build OpenClaw *right now*. The very documentation you are reading was authored collaboratively between a human developer and an AI coding agent.

Part V: Technical Deep Dive

Parts I through IV established *why* an Agent is needed, *what* the LLM brings and lacks, *how* OpenClaw maps to the Agent design pattern, and *what the work of building agents involves*. This part dives into the **implementation specifics**: the codebase structure, the SDK integration, and the configuration-level details that make the pattern concrete.

Codebase Organization

The six Agent layers described in Part III are realized through four directory-level boundaries in the OpenClaw monorepo:

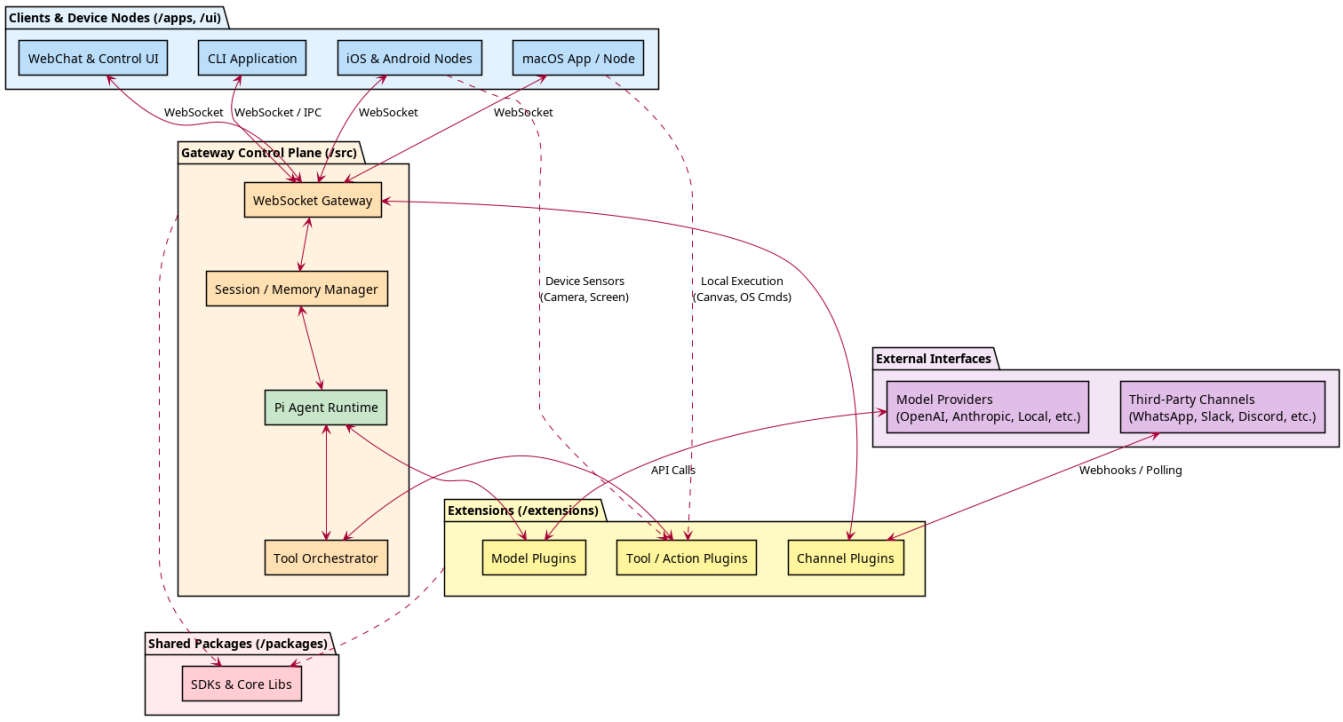
Agent Layer (from Part III)	Codebase Location	What Lives There
Orchestration Layer	`/src`	Gateway Control Plane: WebSocket server, routing engine, session management, tool orchestration, system prompt builder

Agent Layer (from Part III)	Codebase Location	What Lives There
Perception Layer + Brain + Action Layer	`/extensions`	80+ plugins: channel adapters (WhatsApp, Telegram, Discord, Matrix, etc.), LLM providers (Anthropic, OpenAI, DeepSeek, Groq, Ollama, etc.), and tool plugins (browser, web search, media understanding)
Perception Layer (client surfaces)	`/apps` and `/ui`	Web UI (Vite-powered SPA for WebChat and Dashboard), Native Nodes for macOS/iOS/Android (exposing Camera, Microphone, Screen Recording, Canvas UI over WebSocket), and CLI
Shared infrastructure	`/packages`	Reusable SDKs: `clawdbot`, `memory-host-sdk`, `moltbot`

A key architectural decision: the Gateway core (`/src`) stays lean. Domain-specific capabilities (channel protocols, model provider quirks, specialized tools) are pushed into `/extensions` as plugins, keeping the orchestration layer decoupled from the specifics of any single channel, model, or tool.

System Interaction Graph

The following diagram shows how messages flow through these boundaries at runtime:



Pi Agent Runtime: Implementation Details

Part III introduced the **Reasoning Loop** as the layer that implements the Perceive-Think-Act cycle. In OpenClaw, this loop is powered by the **Pi Agent Runtime**, an embedded SDK rather than a standalone subprocess.

OpenClaw imports pi-mono (specifically `pi-coding-agent` and its siblings) directly into the Gateway's Node.js process via `createAgentSession()`. This embedded approach, as opposed to subprocess RPC, gives OpenClaw programmatic control over every phase of the agent lifecycle:

SDK Integration Points

1. **Reasoning Loop (Perceive-Think-Act):** The Pi engine handles the inner turn-by-turn cycle. It sends the assembled prompt to the LLM provider, parses `<think>` reasoning blocks vs `<final>` answers, and fires tool executions dynamically before returning final text to the user. This is the concrete implementation of the iterative loop described in Part I.

2. **Tool Override Pipeline:** As described in Part III ("Addressing Inability to Act"), OpenClaw injects tool definitions into the LLM's prompt. At the implementation level, Pi brings default base tools (read, write, bash), but OpenClaw **replaces all of them**. It strips Pi's default implementations via `splitSdkTools()` and injects its own extended toolset, including browser control, channel-specific actions (e.g., `whatsapp_send`), and companion app sensors. All tools flow through `customTools` so that OpenClaw's policy filtering and sandbox integration apply uniformly.

3. **Session Events Hooking (Real-time Streaming):** OpenClaw subscribes to Pi's `AgentSession` events (`message_start`, `message_update`, `tool_execution_start`, `tool_execution_end`, `turn_end`, `auto_compaction_start`) to translate the AI's internal thought process into user-visible output. This enables real-time streaming updates in the Web UI and chunked block replies to messaging channels.

4. **Compaction Implementation:** As described in Part III ("Addressing Context Limits"), OpenClaw uses two compaction strategies. At the implementation level, Pi's `SessionManager` handles auto-compaction when token context approaches the model's window limit. OpenClaw adds two custom Pi extensions on top: a **Compaction Safeguard** (`compaction-safeguard.ts`) with adaptive token budgeting and tool failure summaries, and **Cache-TTL Context Pruning** (`context-pruning.ts`) that tracks freshness of context items and prunes stale entries.

5. **Dynamic Prompt Assembly:** At session start, the system prompt builder (`buildAgentSystemPrompt()`) reads locally maintained `.md` files (`AGENTS.md`, `SOUL.md`, `TOOLS.md`, `IDENTITY.md`, `USER.md`, and skills via `SKILL.md`) from

`~/.openclaw/workspace/` and assembles them into the system prompt alongside tooling definitions, safety guardrails, and runtime metadata. This is how the "Persistence Layer" from Part III feeds stable context into each new reasoning turn.

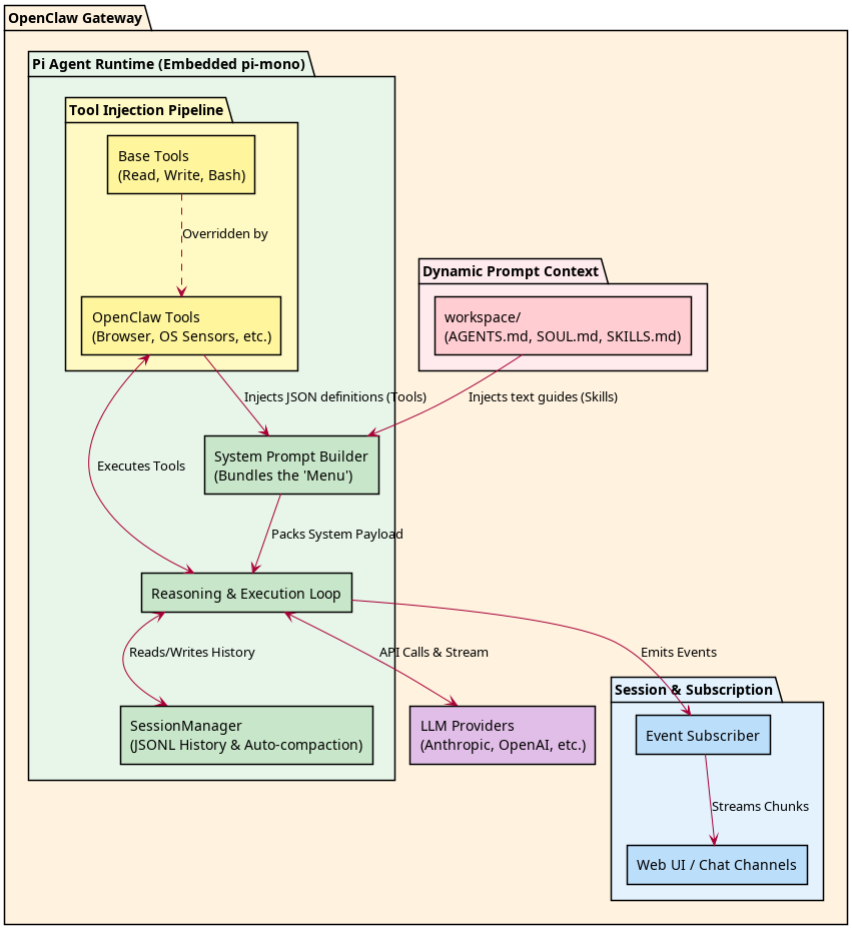
Tooling Responsibility Split: LLM vs Pi

A critical design boundary in any Agent system is: **who decides what to do** vs **who does it**. In OpenClaw, this is cleanly split:

1. **Pi provides the "Menu":** Before each prompt, the Pi runtime bundles JSON Schema definitions for every available tool and sends them in the system prompt. The LLM sees this as: *"Here are the tools you can call, and here are the exact arguments each requires."*
2. **The LLM decides:** Using its reasoning capabilities (a core LLM strength from Part II), the LLM selects a tool and emits a structured tool call request within its response stream.
3. **Pi executes:** Pi intercepts the tool call from the LLM's streaming output, pauses further generation, runs the actual TypeScript backend code, and injects the raw result back into the LLM's context window. The LLM then continues reasoning with the new information. This is the concrete realization of the "observe tool results and feed them back" pattern described in Part II ("The Implication for Agent Design").

This loop repeats within a single user turn until the LLM determines no more tool calls are needed and emits a final response.

Pi Runtime Graph



Security and Execution Model: Implementation Details

Part III introduced the three-tier security model at a conceptual level. Here we cover the **trust model rationale** and **configuration-level specifics** that govern how it operates.

Trust Model

OpenClaw is designed around a **Single-Operator Trust Model ("Personal Assistant")**. The system assumes the person running the Gateway is the trusted owner of the machine. The primary security concern is not protecting the owner from the AI, but preventing **external Prompt Injections** (from third parties messaging the bot through public channels) from escalating into host-level actions.

Configuration Details for the Execution Tiers

The three tiers described in Part III ("Addressing Security") are controlled by specific configuration keys:

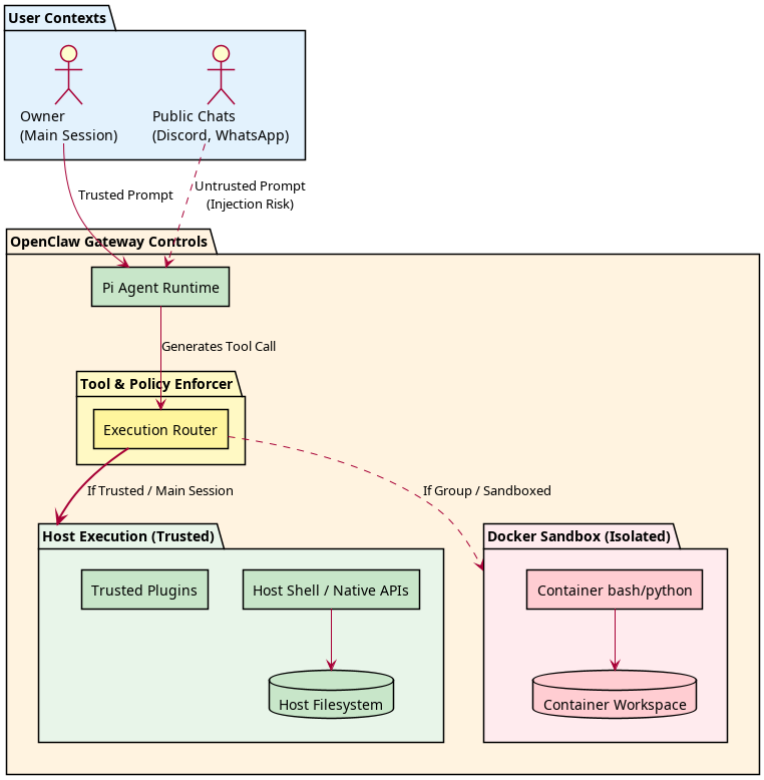
1. **Host-First (Trusted Mode):** The default for the owner's Main Session. Tool executions (`bash`, `python`, `fs`) run directly on the host with the same privileges as the Gateway process. No additional configuration is required.

2. **Sandboxed Mode:** Activated by setting `agents.defaults.sandbox.mode` to `non-main` (sandbox non-owner sessions) or `all` (sandbox everything). When active, local execution tools are routed into a Docker container without host access, isolating untrusted inputs from channels like public Telegram groups or Discord servers.
3. **Restricted Tool Scopes:** Orthogonal to sandbox mode. Setting `tools.fs.workspaceOnly: true` forces all file system tools to operate exclusively within `~/.openclaw/workspace`, regardless of trust level.

Plugin Trust Boundary

One important security nuance not covered in Part III: custom plugins loaded from `/extensions` run **in-process** as part of the trusted computing base. They execute outside the Docker sandbox, with the same host privileges as the Gateway itself. This means installing a malicious or compromised plugin bypasses all sandbox protections. Plugin trust is therefore an operator-level decision, not a runtime-enforced boundary.

Security Boundary Graph



Appendix: Why OpenClaw is Still Needed (Despite Claude Code)

The comparison table in Part III shows that both OpenClaw and Claude Code implement the same 6-layer Agent pattern. If Claude Code already provides a fully functional Agent with

tools, skills, rules, and MCP extensibility, why build OpenClaw?

The answer is that they solve **fundamentally different problems**:

Dimension	Claude Code	OpenClaw
Primary purpose	Coding agent for software engineers	General-purpose personal assistant for anyone
Interaction model	Session-based: you start it, use it, close it	Always-on Gateway: persistent server that listens 24/7
Channels	Terminal and IDE only	WhatsApp, Telegram, Discord, Slack, Web UI, CLI, native apps
Audience	Single developer at a terminal	Multiple users across multiple channels simultaneously
LLM vendor	Anthropic only (vendor lock-in)	Any provider (Anthropic, OpenAI, DeepSeek, Ollama local, etc.)
Device integration	None	Companion apps expose Camera, Microphone, Screen Recording, Notifications, Canvas UI
Domain scope	Code: read, write, edit, execute	Life: messaging, media, smart home, calendar, code, and anything reachable via plugins

In short:

- **Claude Code** is a *coding tool* — you open it when you have a programming task, and close it when done.
- **OpenClaw** is a *personal assistant platform* — it runs in the background, bridges your messaging apps, controls your devices, and happens to also be capable of coding.

The key architectural differentiators are:

1. **Multi-channel presence:** OpenClaw can receive a WhatsApp message, reason about it, execute tools, and reply on WhatsApp — all without the user touching a terminal. Claude Code requires the user to be at a keyboard.
2. **Always-on persistence:** OpenClaw runs as a long-lived server process. It can handle incoming messages at any time, maintain ongoing conversations across sessions, and proactively notify users through their preferred channel.
3. **Provider freedom:** OpenClaw's provider abstraction means you can switch between Claude, GPT-4, DeepSeek, or local Ollama models per session, per channel, or even per message — optimizing for cost, speed, privacy, or capability as needed.
4. **Hardware bridge:** Through companion apps on macOS, iOS, and Android, OpenClaw extends its "hands" to physical device capabilities that no terminal-based agent can

reach.